8-2. Generalised Linear Models (GLMs)
Definition of GLM

A GLM is defined by 3 components:

1. The linear predictor $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$, where $\boldsymbol{\beta}$ is the $p \times 1$ parameter vector and $X = \begin{pmatrix} \mathbf{x}_1^T \\ \vdots \\ \mathbf{x}_n^T \end{pmatrix}$ is the $n \times p$ matrix of explanatory variables.
2. The link function $g(\cdot)$ that relates the mean of the response variable, $\mu_i = \mathsf{E}[Y_i \mid X_i]$, with the linear predictor

$$\begin{aligned} g(\mu_i) &= \eta_i = \mathbf{x}_i^T \boldsymbol{\beta} \\ \Rightarrow \mu_i &= g^{-1}(\eta_i) \end{aligned}$$

3. The random component specifying the distribution of the response variable Y (e.g. normal, Poisson, binomial etc).

Examples

We have already seen the following examples.

- Normal linear regression: $\mu_i = \mathsf{E}[Y_i \mid X_i]$ and
  - $g(\mu_i) = \mu_i = \beta_0 + \beta_1 x_i$ (identity link)
  - $Y_i \sim N(\mu_i, \sigma^2)$
- Poisson trend: $\mu_x = \mathsf{E}[D_x]$ and
  - $g(\mu_x) = \log(\mu_x) = \beta_0 + \beta_1 x$ (log link)
  - $D_x \sim \text{Poisson}(\mu_x)$
- Logistic regression: $\mu_x = \mathsf{E}[\hat{q}_x] = q_x$ and
  - $g(q_x) = \text{logit}(q_x) = \beta_0 + \beta_1 x$ (logit link)
  - $D_x \sim \text{Binomial}(E_x, q_x)$

Definition of GLM continued

- The distribution of the response variable Y must be a member of the exponential family, i.e. have a pdf (or pmf) of the form

$$f(y;\theta) \quad = \quad \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\}$$

  where $a(\cdot), b(\cdot)$ and $c(\cdot,\cdot)$ are functions.

- Parameter θ is called the 'natural parameter' and ϕ is a scale parameter.

Example: Poisson distribution

Poisson(μ) has pmf

$$\begin{aligned} f_Y(y) &= \frac{\mathrm{e}^{-\mu}\mu^y}{y!} \qquad \text{for } y = 0, 1, 2, \ldots \\ &= \exp\left\{\log\left(\frac{\mathrm{e}^{-\mu}\mu^y}{y!}\right)\right\} \\ &= \exp\left\{-\mu + y\log(\mu) - \log(y!)\right\} \\ &= \exp\left\{\frac{y\log\mu - \mu}{1} - \log(y!)\right\} \end{aligned}$$

So the Poisson distribution belongs to the exponential family, with natural parameter $\theta = \log\mu$ and functions $a(\phi) = 1$, $b(\theta) = \mu = \mathrm{e}^\theta$, $c(y, \phi) = -\log(y!)$

Scale parameter ϕ is not needed here, conventional to set $\phi = 1$.

Example: Normal distribution

$N(\mu, \sigma^2)$ pdf is

$$\begin{aligned} f_Y(y) &= \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2\right\} \\ &= \exp\left\{\frac{\mu y - \frac{1}{2}\mu^2}{\sigma^2} + \log\left(\frac{1}{\sigma\sqrt{2\pi}}\right) - \frac{y^2}{2\sigma^2}\right\} \end{aligned}$$

So the Normal distribution belongs to the exponential family, with natural parameter $\theta = \mu$, scale parameter $\phi = \sigma^2$, and functions $a(\phi) = \phi$, $b(\theta) = \frac{1}{2}\theta^2$, $c(y, \phi) = \log\left(\frac{1}{\sqrt{2\pi\phi}}\right) - \frac{y^2}{2\phi}$

Exercise: check that Binomial distribution belongs to the exponential family.

- The functions $a(\phi)$, $b(\theta)$, $c(y, \phi)$ are not uniquely defined. For a distribution to belong to the exponential family, we just need some such functions to exist.

- If the function defining the natural parameter θ is used as the link function in a GLM, then the link is known as the **canonical link**, e.g.
  - $N(\mu, \sigma^2)$: natural parameter is mean μ
    canonical link is $g(\mu) = \mu$ (identity)
  - Poisson(μ): natural parameter is $\log(\mu)$
    canonical link is $g(\mu) = \log(\mu)$ (log link)
  - Binomial(E, q): natural parameter is $\log\left(\frac{q}{1-q}\right)$
    canonical link is $g(q) = \log\left(\frac{q}{1-q}\right)$ (logit link)

  Note: the canonical link is just an algebraic convenience, it's fine to use other link functions.